

3.1 An Introduction to AS Level Organic Chemistry

Question Paper

Course	CIE A Level Chemistry
Section	3. Organic Chemistry
Topic	3.1 An Introduction to AS Level Organic Chemistry
Difficulty	Hard

Time allowed: 30

Score: /25

Percentage: /100

Question 1a

This question is about organic compounds found in living organisms.

Terpenes are a class of organic compounds widely used as fragrances in products such as perfumes. For example, one of the simpler terpenes, nerol is found in many essential oils such as lemongrass and hops. The structure of nerol is shown in **Fig 1.1**.

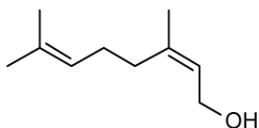


Fig 1.1

i)

State the molecular formula of nerol and calculate the relative molecular mass of nerol.

[2]

ii)

Draw the displayed formula of nerol.

[1]

[3 marks]

Question 1b

Explain whether nerol shows geometrical (cis-trans) and optical isomerism. In your answer, identify any relevant points in the structure of nerol.

[3 marks]

Question 1c

Two structural isomers of nerol are linalool and geraniol. They are unsaturated alcohols with similar chemical properties that are both used in perfumes.

The IUPAC names for both linalool and geraniol are shown in **Table 1.1**.

Table 1.1

Isomer	IUPAC name
Linalool	3,7-dimethylocta-1,6-dien-3-ol
Geraniol	3,7-dimethylocta-2,6-dien-1-ol

Draw the skeletal formulae for these two isomeric alcohols.

[2 marks]

Question 2a

The organic compounds in Fig 2.1 are all made by living organisms.

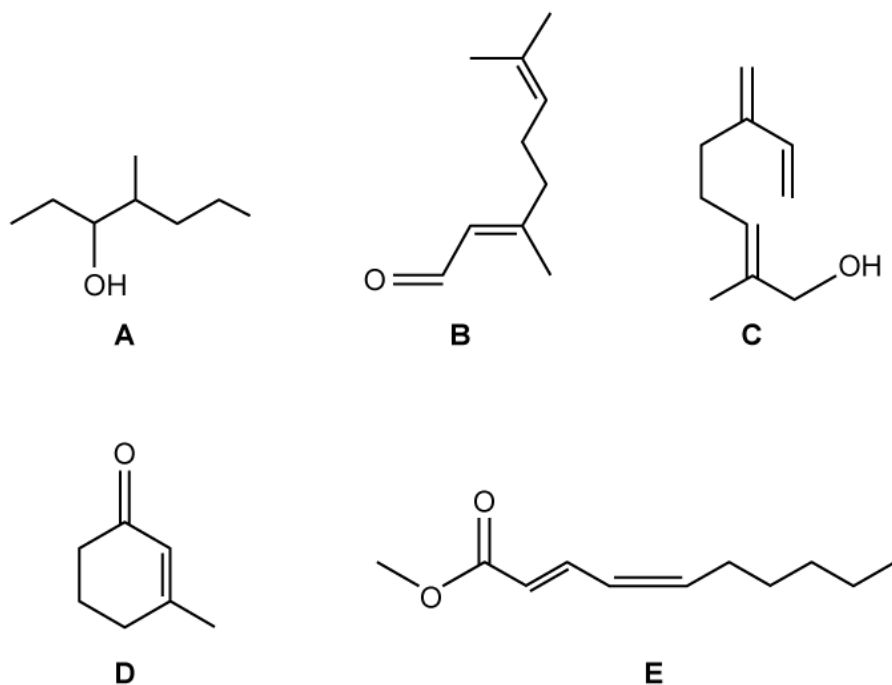


Fig. 2.1

i)

Give the name of compound **A**.

[1]

ii)

Compound **A** has two chiral centres and forms stereoisomers.

Draw **one pair** of optical isomers of compound **A** using the conventional three-dimensional representation.

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[2]

[3 marks]

Question 2b

Compounds **B**, **C**, **D** and **E** all show the same type of isomerism.

Circle the functional group(s) responsible for this isomerism on the molecules in Fig 2.1.

[3 marks]

Question 2c

Compound **D** reacts with hydrogen chloride to form two products.

i)

Fully name the type of mechanism for this reaction.

[1]

ii)

One of the products formed exists as a pair of stereoisomers.

Draw three-dimensional diagrams of the pair of stereoisomers formed during this reaction.

Indicate with an asterisk (*) the chiral centre on one of the structures drawn.

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[3]

[4 marks]

Question 2d

Name the type of reaction occurring when compound **E** reacts to form methanol and deca-2,4-dienoic acid.

[1 mark]

Question 3a

Ibuprofen is a common non-steroidal anti-inflammatory drug (NSAID). It contains a benzene ring and a carboxylic acid at the end of one of the branches as shown in Fig 3.1.

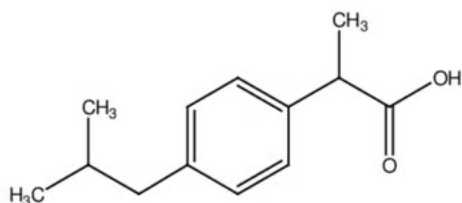


Fig 3.1

Deduce the number of sigma (σ) and pi (π) bonds in ibuprofen.

sigma

pi

[2 marks]

Question 3b

The ibuprofen molecule contains both sp^2 and sp^3 hybridised carbon atoms.

State the number of sp^2 and sp^3 hybridised carbon atoms in a molecule of ibuprofen.

[2 marks]

Question 3c

Explain why the benzene ring is a planar hexagon.

[2 marks]

Model Answers: Hard

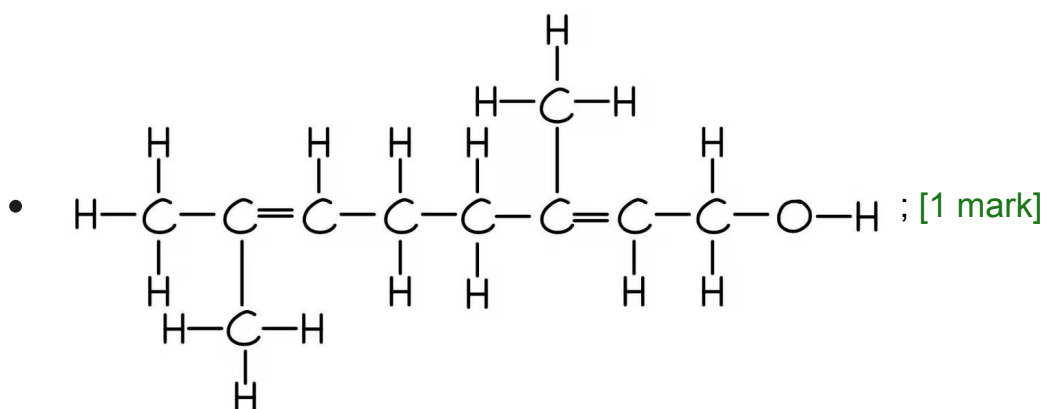
1a

a)

i) The molecular formula and relative molecular mass of nerol are:

- $C_{10}H_{18}O$; [1 mark]
- 154.0; [1 mark]

ii) The displayed formula of geraniol is:



[Total: 3 marks]

- The molecular formula tells you the number of each type of atom present in the molecule
 - It is a case of counting out each type of atom present but it involves you taking your time, especially when counting up the hydrogens present
- The molar mass calculation and drawing of the displayed formula should be relatively straightforward once you have got the molecular formula correct

1b

b) Explaining whether nerol show geometrical (cis-trans) or optical isomerism:

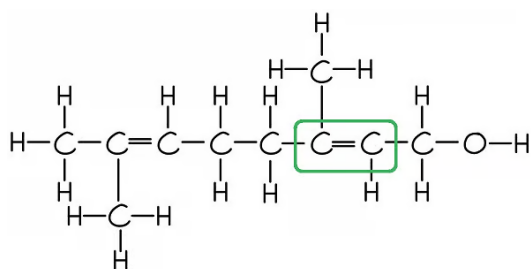
Geometrical (cis-trans):

- Nerol does show / exhibit geometrical (cis-trans) isomerism

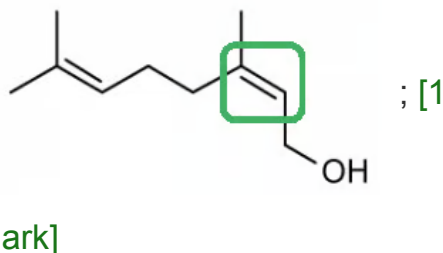
AND

Because there are four different groups around one of the carbon=carbon double bonds / C=C; [1 mark]

- Correct carbon=carbon double bond identified, e.g.



OR



Optical:

- Nerol does not show / exhibit optical isomerism

AND

Because there are no chiral carbons / centres **OR** because there are no carbons that have four different atoms / groups of atoms attached; [1 mark]

[Total: 3 marks]

- Geometrical (cis-trans) isomers require a carbon=carbon double bond / C=C **AND** a minimum of two different groups on either side of the carbon=carbon double bond / C=C
 - Nerol contains two carbon=carbon double bonds / C=C
 - The first one has two methyl groups attached to the left-hand carbon
 - So, this carbon=carbon double bond / C=C cannot show / exhibit geometrical (cis-trans) isomerism
 - The second one has a methyl group and a large R group attached to the left-hand carbon and a hydrogen atom and a CH₂OH group attached to the right-hand carbon
 - So, this carbon=carbon double bond / C=C can show / exhibit geometrical (cis-trans) isomerism
- Optical isomerism requires a chiral carbon / centre
 - Remember:** A chiral carbon is a carbon atom that has four different atoms of

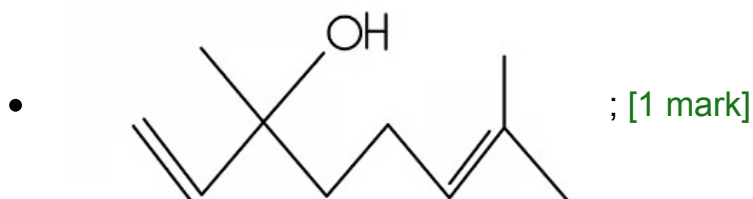
groups of atoms bonded to it

- There are no chiral carbons / centres in nerol

1c

c) The skeletal formulae for linalool and geraniol are:

Linalool / 3,7-dimethylocta-1,6-dien-3-ol:



Geraniol / 3,7-dimethylocta-2,6-dien-1-ol



[Total: 2 marks]

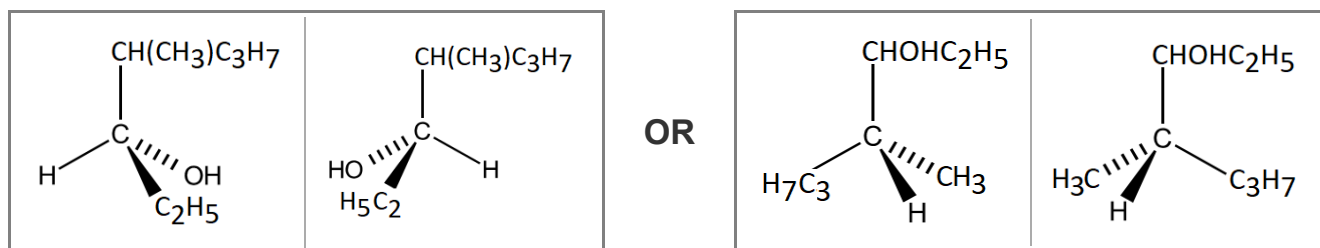
- Drawing the skeletal formulas here requires working through the IUPAC name methodically
- They are unsaturated alcohols as it says in the question stem:
 - Octa means the longest carbon chain is 8 carbons long
 - Alcohol is the higher priority group and IUPAC rules state that the numbering should be as low as possible, meaning:
 - C-OH is carbon 3 (Linalool)
 - C-OH is carbon 1 (Geraniol)
 - 3,7-dimethyl means there are two methyl (-CH_3) groups at carbons 3 and 7 for both Linalool and Geraniol
 - 1,6-dien (Linalool) indicates that there are two double bonds at carbon 1 and carbon 6
 - 2,6-dien (Geraniol) indicates that there are two double bonds at carbon 2 and carbon 6

2a

a)

i) The name of compound **A** is:

- 4-methylheptan-3-ol; [1 mark]

ii) **One pair** of optical isomers of compound **A** using the conventional three-dimensional representation is:

- One correct 3-D structure; [1 mark]
- A mirror image of the first molecule; [1 mark]

[Total: 3 marks]

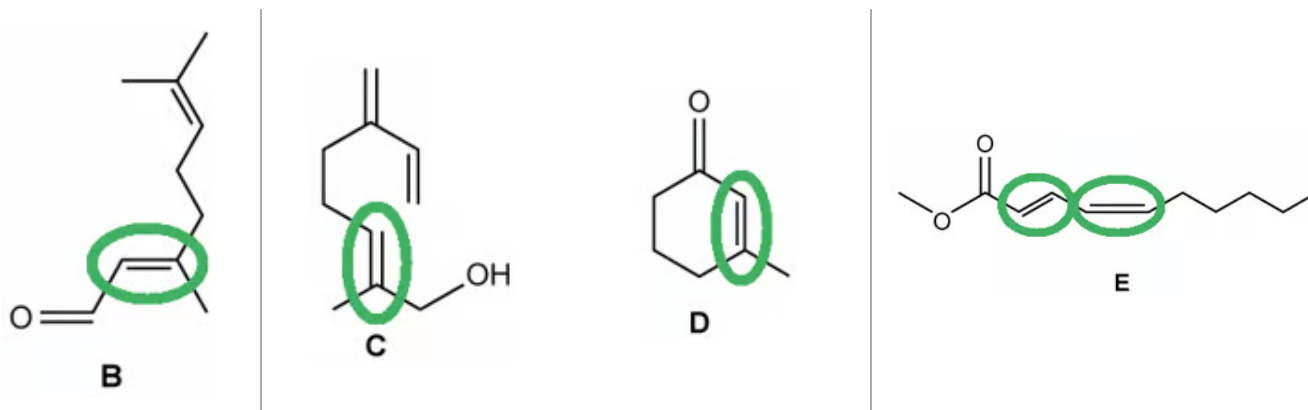
- For part (i)
 - The longest carbon chain is 7 carbons long with only single bonds
 - This will provide -heptan- to the name
 - There is an OH group on carbon 3
 - The OH group takes higher priority in nomenclature
 - This will provide -3-ol to the name
 - There is a CH₃ group on carbon 4
 - This will provide 4-methyl to the name
 - These pieces combine to give 4-methylheptan-3-ol
- For part (ii)
 - **Remember:** Chiral centres have four different atoms or groups of atoms
 - You are told that there are two chiral centres
 - These are found on carbon-3 and carbon-4 of the carbon chain
 - The different groups attached to carbon-3 are:
 - OH group
 - H atom
 - Ethyl / C₂H₅ group
 - R group - take care with this R group as there is a "hidden" hydrogen on

carbon-4 as well as the methyl / CH_3 group

- The different groups attached to carbon-4 are:
 - Methyl / CH_3 group
 - H atom
 - Propyl / C_3H_7 group
 - R group - take care with this R group as there is a "hidden" hydrogen on carbon-3 as well as the OH group
- Instead of drawing mirror images, you can achieve the second mark by drawing the same molecule but with two of the functional groups swapped

2b

b) The functional group(s) responsible for the similar isomerism in compounds **B**, **C**, **D** and **E** are:



Compound **B**:

- Carbon=carbon double bond / $\text{C}=\text{C}$ highlighted as shown
AND
No other carbon=carbon double bond / $\text{C}=\text{C}$ highlighted; [1 mark]

Compound **C**:

geometrical (cis-trans) isomerism

- **Compound B:**
 - The top carbon=carbon double bond / C=C has two methyl groups attached to the same carbon atom
 - Therefore, this carbon=carbon double bond / C=C cannot show / exhibit geometrical (cis-trans) isomerism
 - The bottom carbon=carbon double bond / C=C has different groups attached to both carbon atoms
 - Therefore, this carbon=carbon double bond / C=C can show / exhibit geometrical (cis-trans) isomerism
- **Compound C:**
 - The top two carbon=carbon double bonds / C=C both have two hydrogen atoms attached to one carbon atom
 - Therefore, these carbon=carbon double bonds / C=C cannot show / exhibit geometrical (cis-trans) isomerism
 - The bottom carbon=carbon double bond / C=C has different groups attached to both carbon atoms
 - Therefore, this carbon=carbon double bond / C=C can show / exhibit geometrical (cis-trans) isomerism
- **Compound D:**
 - The only carbon=carbon double bond / C=C has different groups attached to both carbon atoms
 - Therefore, this carbon=carbon double bond / C=C can show / exhibit geometrical (cis-trans) isomerism
- **Compound E:**
 - Both carbon=carbon double bonds / C=C in the chain have different atoms / group attached to the carbon atoms of the bond
 - Therefore, both carbon=carbon double bonds / C=C can show / exhibit geometrical (cis-trans) isomerism

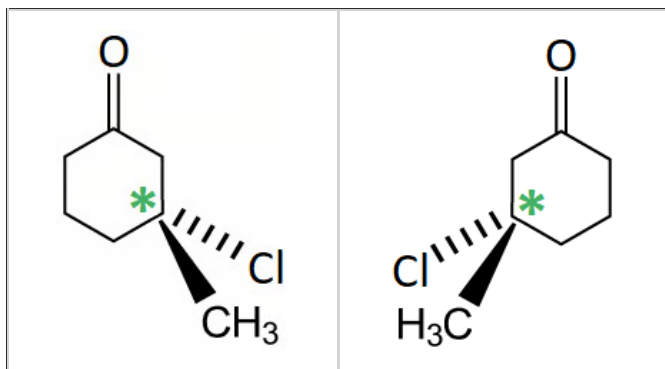
2c

c)

i) When compound **D** reacts with hydrogen chloride, the type of mechanism is:

- Electrophilic addition; [1 mark]

ii) The three-dimensional diagrams of the pair of stereoisomers formed during this reaction are:



- One correct 3-D structure; [1 mark]
- A mirror image of the first molecule; [1 mark]
- A chiral C shown as * ; [1 mark]

[Total: 4 marks]

- For part (i)
 - The presence of a carbon=carbon double bond / C=C should indicate a potential electrophilic addition reaction / mechanism
 - The second reagent being hydrogen chloride should confirm this
 - The hydrogen chloride adds across the double bond to form two possible products following Markovnikov's rule
 - 2-chloro-3-methylcyclohexanone is the minor product
 - 3-chloro-3-methylcyclohexanone is the major product as there is the ring structure and the methyl group to stabilise the carbocation intermediate
 - **Remember:** Chiral centres have four different atoms / groups of atoms
 - 3-chloro-3-methylcyclohexanone has four different atoms / groups of atoms
 - The ring structure moving clockwise towards the carbonyl group
 - The ring structure moving anti-clockwise towards the carbonyl group
 - The chlorine atom
 - The methyl / CH₃ group
 - Chiral centres within ring systems are often more challenging to identify / draw
 - **Tip:** Use the two bonds that are in the same plane (not the dotted or wedge bond) to draw the ring structure then add the other two atoms / groups of

atoms

- Instead of drawing mirror images, you can achieve the second mark by drawing the same molecule but with the chlorine and methyl / CH₃ group swapped
- For the final mark, make sure that you put an asterisk (*) on at least one of the chiral carbons

2d

d) The type of reaction occurring when compound **E** reacts to form methanol and deca-2,4-dienoic acid is:

- Hydrolysis; [1 mark]

[Total: 1 mark]

- Compound **E** is an ester
- The reaction to form methanol and deca-2,4-dienoic acid is the breakdown of an ester to form its parent alcohol and carboxylic acid
- This can be done under different conditions but the overall reaction is hydrolysis
 - The carbon-oxygen single bond / C-O of the ester linkage has been broken
 - Hydrogen is added to the single-bonded oxygen to form methanol
 - An OH group has been added to the carbonyl carbon to form deca-2,4-dienoic acid
 - Overall, water (hydro) has been added and compound **E** has been broken down (lysis)

3a

a) The number of sigma (σ) bonds and pi (π) bonds in ibuprofen is:

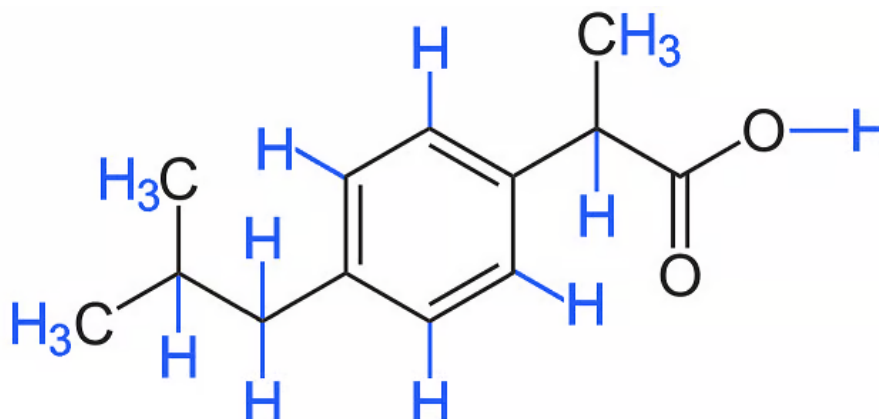
- Sigma = 33; [1 mark]
- Pi = 4; [1 mark]

[Total: 2 marks]

- **Careful:** It is very easy to miss sigma bonds in the ibuprofen structure because:
 - Some have been written into the CH₃ and OH groups
 - There are 8 "hidden" hydrogens in the structure
 - 4 in the chain portions of the molecule

- 4 in the ring structure of the molecule

- **Tip:** Re-draw the molecule in a way that includes the "hidden" hydrogen atoms AND highlights the bonds that are within CH₃ and OH groups



- **Remember:** All covalent bonds are sigma bonds
 - Single bonds are sigma bonds
 - Double bonds are one sigma and one pi bond
 - Triple bonds are one sigma and two pi bonds
- There are 33 bonds in total
 - Therefore, there are 33 sigma bonds
- There are 4 double bonds
 - Therefore, there are 4 pi bonds

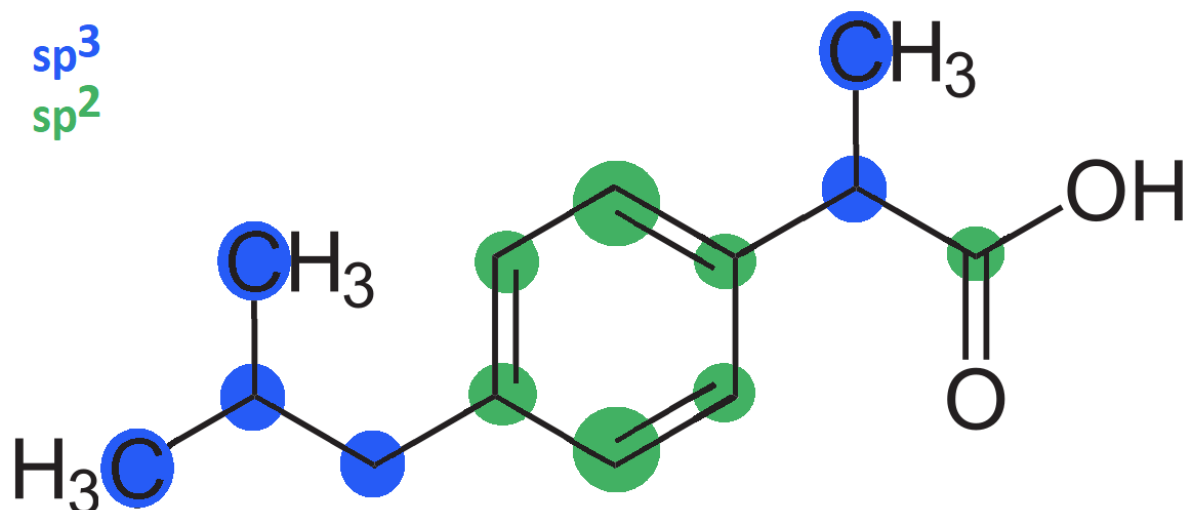
3b

b) The number of sp² and sp³ hybridised carbon atoms in a molecule of ibuprofen is:

- sp² = 7; [1 mark]
- sp³ = 6; [1 mark]

[Total: 2 marks]

- Each of the carbons with a tetrahedral geometry has sp³ hybridised orbitals
- Each of the carbons in the benzene ring, plus the carbonyl carbon have sp² hybridised orbitals



3c

c) The shape of the benzene ring is a regular, planar hexagon because:

- Planar = All of the carbon atoms are sp^2 hybridised
OR
All of the carbon atoms have trigonal planar geometry
OR
All of the carbon atoms have bond angles of 120° ; [1 mark]
- Hexagonal = There are six carbons; [1 mark]

[Total: 2 marks]

- All of the carbons in the benzene ring are sp^2 hybridised which means that their shape is trigonal planar
 - This causes the "planar" shape of benzene
- Since the benzene ring contains 6 bonded carbons with no other atoms in-between them, its shape must be a hexagon